

Participant Engagement and Data Compliance in Remote Digital Studies for ADHD, Autism, and Eating Disorders

Heet Sankesara

heet.sankesara@kcl.ac.uk

Institute of Psychiatry, Psychology &
Neuroscience King's College London
United Kingdom

Pauline Conde

pauline.conde@kcl.ac.uk

Institute of Psychiatry, Psychology &
Neuroscience King's College London
United Kingdom

Yatharth Ranjan

yatharth.ranjan@kcl.ac.uk

Institute of Psychiatry, Psychology &
Neuroscience King's College London
United Kingdom

Isabel Yorke

isabel.d.yorke@kcl.ac.uk

Institute of Psychiatry, Psychology &
Neuroscience King's College London
United Kingdom

Alice Barnes

alice.e.barnes@kcl.ac.uk

Institute of Psychiatry, Psychology &
Neuroscience King's College London
United Kingdom

Ulrike Schmidt

ulrike.schmidt@kcl.ac.uk

Institute of Psychiatry, Psychology &
Neuroscience King's College London
United Kingdom

Richard JB Dobson

richard.j.dobson@kcl.ac.uk

Institute of Psychiatry, Psychology &
Neuroscience King's College London
United Kingdom

Amos A Folarin

amos.folarin@kcl.ac.uk

Institute of Psychiatry, Psychology &
Neuroscience King's College London
United Kingdom

Zulqarnain Rashid

zulqarnain.rashid@kcl.ac.uk

Institute of Psychiatry, Psychology &
Neuroscience King's College London
United Kingdom

Abstract

The rapid advancement of digital technologies has facilitated the recruitment and remote monitoring of large, diverse populations in health research. However, inconsistent participant engagement and attrition over time can significantly compromise the reliability and generalisability of insights derived from remotely collected health data. We present findings on long-term retention and engagement patterns from large-scale, multinational observational digital studies on three conditions: Attention-Deficit/Hyperactivity Disorder (ADHD), Autism, and Eating Disorders. The studies involved 1200 participants; we collected both active data (scheduled surveys) and passive sensor data via Android smartphones and wearable devices, including Fitbit, Empatica EmbracePlus and Oura Ring, over a period of up to two Years. Our results show that a shorter study period, ease of managing wearable devices, and consistent participant engagement substantially enhance the quantity and quality of collected data.

CCS Concepts

• **Human-centered computing** → **Smartphones; Mobile devices**; • **Applied computing** → **Health informatics**.

Keywords

mHealth, Wearable Devices, Smartphones, Data Collection Platform, Mental Health, Engagement, Retention

ACM Reference Format:

Heet Sankesara, Pauline Conde, Yatharth Ranjan, Isabel Yorke, Alice Barnes, Ulrike Schmidt, Richard JB Dobson, Amos A Folarin, and Zulqarnain Rashid. 2025. Participant Engagement and Data Compliance in Remote Digital Studies for ADHD, Autism, and Eating Disorders. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3755880>

1 Introduction

Mobile health (m-health) data monitoring offers a new approach to obtaining more detailed types and frequent longitudinal behavioural data in real-world settings. Active monitoring data involves collecting with direct user interaction, e.g. a questionnaire via a smartphone app when a user receives a notification. Passive monitoring using smartphones and wearable sensors enables ongoing, unobtrusive data collection while the participant continues with their everyday activities; usefully, these data are more objective as they are collected without the conscious input of the user. The real-world behaviours that can be captured using such passive monitoring may, for example, provide digital signals of symptom levels or predict outcomes. mHealth data collection platforms like [13] made it possible to recruit thousands of participants within just a few months, facilitating large-scale collection of real-world data at significantly lower costs than traditional methods. However, a major challenge remains: high participant attrition rates in remote studies. This raises concerns about the representativeness of the data and the generalisability of research findings.

This work will explore participants' engagement and retention over time from three different mental disorder studies. We have used data availability as a proxy for engagement and compliance. Studies include ADHD Remote Technology study of cardio-metabolic risk



This work is licensed under a Creative Commons Attribution 4.0 International License.
UbiComp Companion '25, Espoo, Finland
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3755880>

factors and medication adherence (ART-CARMA), Eating Disorders: Delineating illness and recovery trajectories to inform personalised prevention and early intervention in young people (EDIFY) and Autism Innovative Medicine Studies-2-Trials (AIMS-2-TRIALS). ART-CARMA is a multi-site prospective cohort study within the EU-funded collaboration TIMESpan, which aims to monitor adults with ADHD. AIMS-2-TRIALS explores how autism develops from before birth to adulthood and how this varies in different people. STORY is a multi-site prospective cohort study within the 4-year interdisciplinary EDIFY research programme that addresses gaps in knowledge of eating disorder risk in young people.

We investigated three key questions using participant-level usage data from study apps and wearables: (i) Is participant retention influenced by real-world factors such as mode of data collection (e.g., smartphones vs. wearables) and devices used? (ii) Is participant engagement affected by study duration, i.e., do long-term studies have higher attrition rates? (iii) Can participant engagement improve retention in a long-term study?

Our findings could inform the design of future digital health studies to enable equitable and balanced data collection from diverse populations.

2 Related Work

Numerous studies have collected longitudinal data and reported participant engagement in remote digital health research [10] [11]. The study Indicators of retention in remote digital health studies: a cross-study evaluation of 100,000 participants [8] emphasized the importance of recruiting a representative population and diverse data samples to enhance engagement. Maximizing Engagement in Mobile Health Studies: Lessons Learned and Future Directions [3] highlighted several strategies to boost engagement, including establishing a clear point of contact with participants. The systematic review Digital health tools for the passive monitoring of depression [1] recommended improved reporting standards, particularly regarding generalisability and reproducibility, including the use of diverse samples, detailed methodologies, and identification of potential biases. Long-term participant retention and engagement patterns in an app and wearable-based multinational remote digital depression study [14] examined long-term engagement and retention trends using data from the RADAR-CNS project. A study on elderly populations, Understanding factors influencing the adoption of mHealth by older adults: An extension of the UTAUT model [5], investigated engagement patterns in this demographic. Remote Assessment of Disease and Relapse in Major Depressive Disorder (RADAR-MDD) [7] provided insights into recruitment, retention, and data availability in a longitudinal remote monitoring study involving individuals with depression. RADAR-base: Major Depressive Disorder and Epilepsy Case Studies [12] assessed the completeness and quality of remote data collected from individuals with epilepsy and major depressive disorder. This work examines three distinct cohorts across different age groups to investigate their engagement and retention patterns.

3 Methods and Procedures

Three distinct remote monitoring studies (ART-CARMA, AIMS-2-TRIALS, STORY) employed mixed-method approaches using the

RADAR-base platform, integrating active (questionnaires, tasks) and passive (wearables, smartphone sensors) data collection.

3.1 RADAR-base Data Collection Platform

To facilitate the wearables and smartphone data remote collection at scale and digital biomarkers development, the RADAR-base platform was released under the open-source Apache 2 licence in January 2018 [9]. RADAR-base platform comprises an Apache Kafka-based back-end deployed onto Kubernetes infrastructure and two mobile apps. The cross-platform (Android, iOS) Cordova Active Remote Monitoring App (aRMT) for active monitoring of participants, requiring conscious action (e.g. questionnaires, audio questions, timed tests, etc.), and the Passive Remote Monitoring App (pRMT), native Android app for passive monitoring of participants via phone, wearable devices and IoT.

3.2 ART-CARMA Study Protocol

The ART-CARMA study recruited 305 adults with ADHD from clinics in the UK and Spain over 24 months, monitoring participants for 12 months before and after ADHD medication initiation[2]. ART-CARMA deployed daily medication questionnaires, weekly side-effect reports (CADDRA), and monthly symptom surveys (e.g., ADHD diagnosis (BAARS-IV), depression (PHQ-8), anxiety (GAD-7)) alongside monthly speech tasks. ART-CARMA used smartphone sensors through the pRMT and Empatica Embraceplus smartwatch to collect the passive data. In a year, participants in the study are scheduled to do 564 questionnaires. In this work, we have looked at the data from the clinics in the UK (N=153).

3.3 AIMS-2-TRIALS Study Protocol

The study administered a daily sleep quality item, biweekly ADHD ratings, and fortnightly social communication questionnaire (SRS) via the RADAR aRMT app. At the end of the 28 days, system usability and user experience questionnaires are scheduled. Moreover, Fitbit Inspire 2 wristbands are given to participants to monitor longitudinal physiological data. 90 participants (both autistic and non-autistic participants combined) have been enrolled for the remote measurement protocol, but we have only used the data from the autistic cohort (N=40). During the monitoring period of 28 days, participants are scheduled to do 46 questionnaires in total.

3.4 STORY Study Protocol

The STORY cohort is planning to include 840 youth (720 with anorexia nervosa or bulimic-type eating disorders, 120 healthy controls) to be recruited via early intervention services and social media, with monitoring spanning 12 months[4, 6]. STORY included biweekly eating disorder symptom surveys, 6-weekly experience sampling (ESM) for mood/stress, and quarterly psychological, cognitive, and neuroimaging assessments. The study used smartphone sensors through the RADAR-base Passive App and Oura ring to collect passive psychological data. Around 264 participants have been recruited to date, and the participants are scheduled to do 301 questionnaires during a year-long monitoring period.

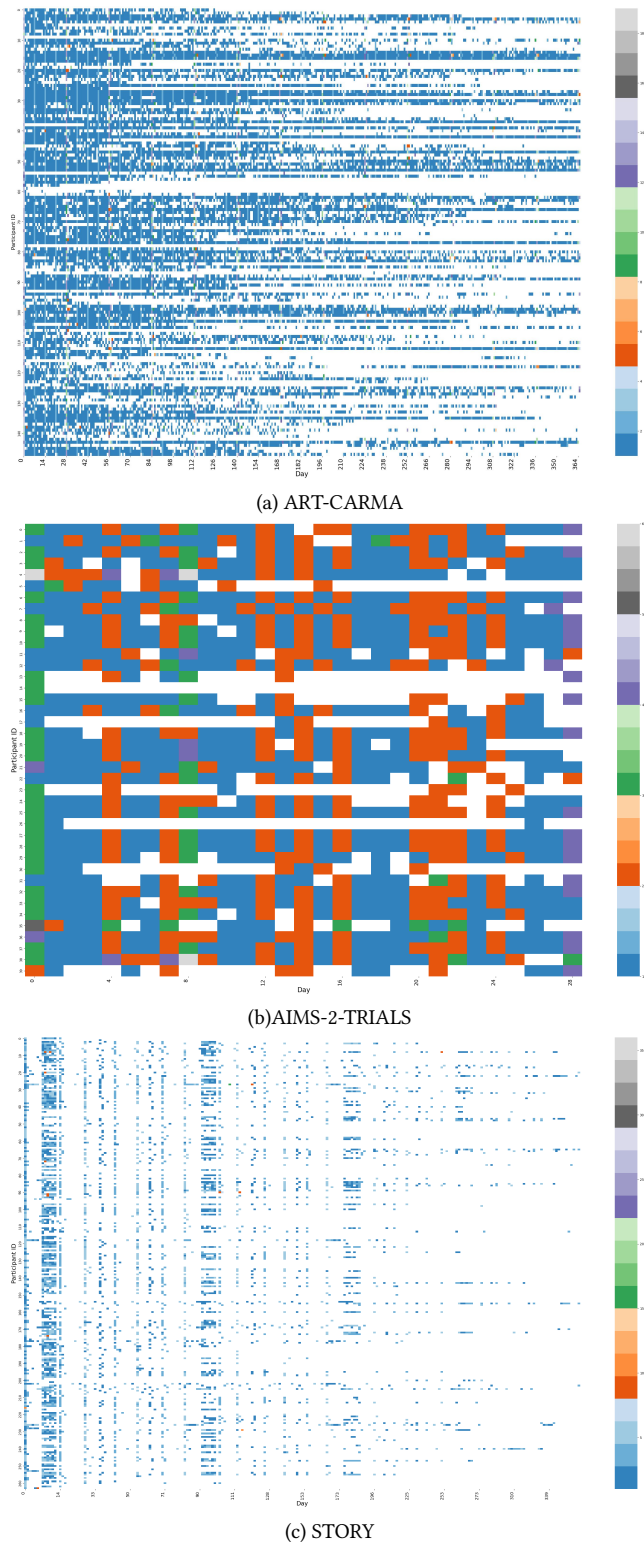


Figure 1: The figures depict the engagement with the scheduled questionnaires via the aRMT app of all the participants over the study period for (a) ART-CARMA, (b) AIMS-2-TRIALS and (c) EDIFY studies. The X-axis shows the days since they joined the study, and the Y-axis shows the participant ID. The colour scale represents the number of questionnaires completed on each day.

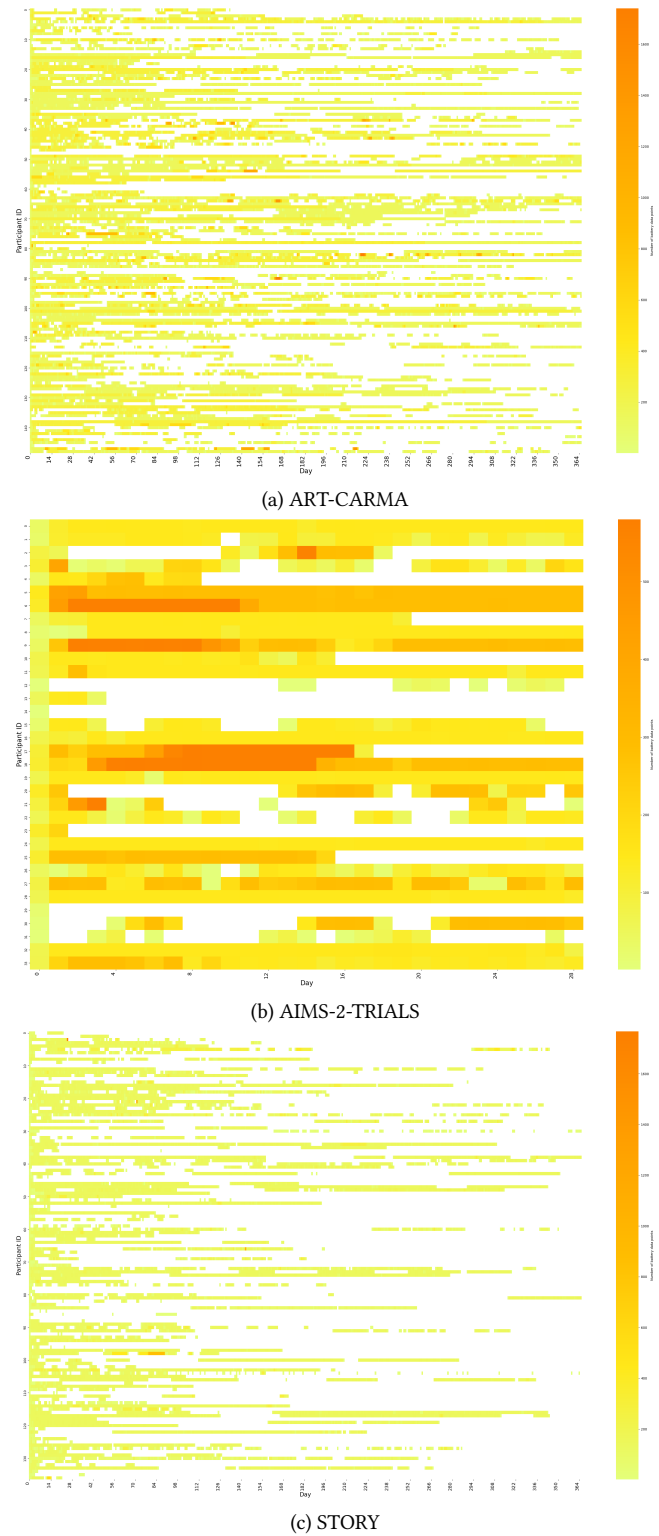


Figure 2: The figures depict the pRMT compliance of all the participants from the (a) ART-CARMA, (b) AIMS-2-TRIALS and (c) STORY studies. The X-axis shows the days since they joined the study and y-axis shows the participant ID. The colour scale shows the number of battery data points collected by the app on a day.

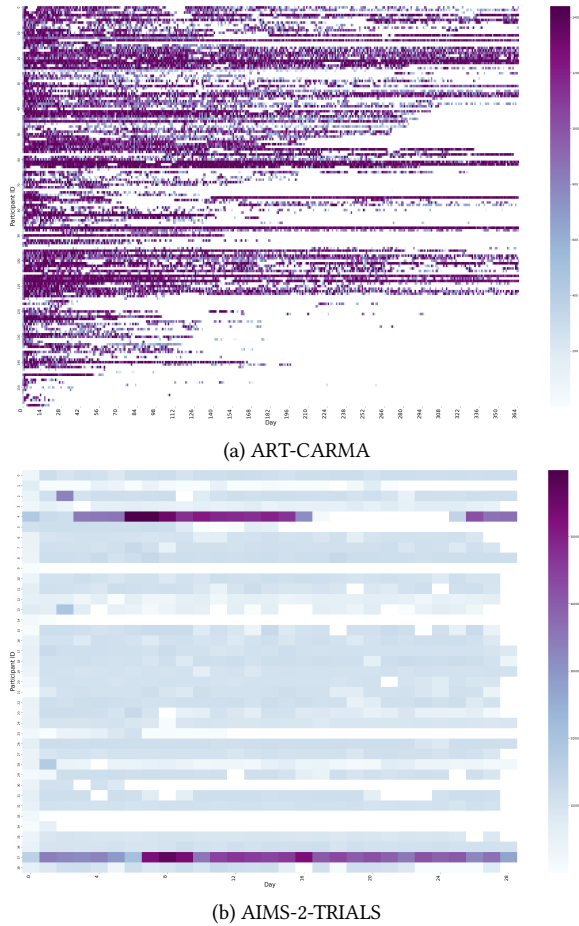


Figure 3: The figures illustrate participant engagement with wearable devices in (a) ART-CARMA and (b) AIMS-2-TRIALS. The X-axis shows the days since they joined the study, and the Y-axis shows the participant ID. The colour scale represents the total daily wear time for (a) and the number of intraday heart data points the wearable collected daily for (b).

3.5 Long term Engagement

In this work, we have done a preliminary analysis of engagement across all three studies for data collected via the aRMT app, pRMT app and wearables. For the aRMT app, we calculated the ratio of questionnaires completed by the participants to the total number of questionnaires scheduled to date. For the pRMT data, we computed the ratio of days when participants have at least one phone battery data point over their total days in the study. To calculate the engagement for the wearable data in the AIMS-2-TRIALS study, we computed the days with at least one intraday heart rate data point from Fitbit over their total days in the study. For ART-CARMA, we have used the digital marker provided by Empatica called wearing detection. For each participant, we calculated days with at least a minute for which the device was worn correctly over the total days in the study.

4 Results

Since the ART-CARMA and STORY studies are ongoing, we looked at the data till 26th March 2025. The ART-CARMA study has recruited 153 participants in London. The aRMT compliance for the participants is 50.06% (figure 1a), the pRMT compliance is 46.50% (figure 2a) and the average wearable compliance is 51.06% (figure 3a) as can be seen in table 1. We have data from all the participants for all the modalities in the study.

For the AIMS-2-TRIALS study, the aRMT compliance is 75.60% for 40 participants (figure 1b), the pRMT compliance is 71.21% for 34 participants (figure 2b) and the wearable compliance is 81.86% for 39 participants (figure 3b).

For the STORY study, the aRMT compliance is 42.3% for 264 participants (figure 1c), and pRMT compliance is 36.90% for 137 participants (figure 2c). We do not calculate their wearable compliance as the wearable component was optional.

Table 1: Data Completion Rate for all Three Studies

Protocol	ART-CARMA	AIMS-2-Trials	STORY
aRMT	50.06%	75.60%	42.3%
pRMT	46.50%	71.21%	36.90%
Wearable Device	51.06%	81.86%	-

The results provide insights into retention and engagement patterns in remote monitoring studies. Our analysis confirms that data collection modalities and device reliability directly shape participant retention: wearables like Fitbit demonstrated higher compliance than the pRMT app, which suffered from Android exclusivity and battery drain. By contrast, the Empatica EmbracePlus exhibited severe attrition due to connectivity failures, battery degradation beyond its 2-year lifespan, and frequent replacements. Our results show that study duration intrinsically elevates attrition risk, as seen in declining compliance curves across long-term studies (Figs. 1,2,3). However, proactive engagement can counteract long-term attrition. The ART-CARMA maintained high compliance over 12 months through researcher-participant interactions despite its intensive protocol, confirming engagement’s power to sustain participation.

5 Conclusion and Future Works

Our findings show that improved data collection in remote digital healthcare studies depends on three key factors: (1) Simplifying study protocols and ensuring that study duration is kept to only what is necessary may help reduce participant burden and minimise dropout rates. (2) ensuring wearables are easy to use, comfortable, and low-maintenance could support consistent usage; (3) maintaining regular, meaningful participant engagement may boost involvement and accountability. These elements collectively enhance data completeness, reliability, and quality. In future work, we will incorporate participant feedback into our analysis. We recognise that the methodology employed in this work to assess engagement for pRMT and wearables may establish a relatively low benchmark for engagement. As we continue to develop this research, we aim to enhance our methodology to ensure a more robust assessment.

Acknowledgments

Authors receive funding from the Medical Research Council/Arts and Humanities Research Council/Economic and Social Research Council Adolescence, Mental Health and the Developing Mind initiative as part of the STORY programme (grant number MR/W002418/1). The results leading to this publication have received funding from the Innovative Medicines Initiative 2 Joint Undertaking under grant agreement No 777394 for the project AIMS-2-TRIALS. This Joint Undertaking receives support from the European Union's Horizon 2020 research and innovation programme and EFPIA and AUTISM SPEAKS, Autistica, SFARI. Any views expressed are those of the author(s) and not necessarily those of the funders (IHI-JU2). ART-CARMA study is part of the TIMESPAN project that has received funding from the European Union's Horizon 2020 research and innovation program under grant agreement No 965381. This reflects only the author's view, and the European Commission is not responsible for any use that may be made of the information it contains.

References

- [1] Lewis S. White K. et al. De Angel, V. 2022. Digital health tools for the passive monitoring of depression: a systematic review of methods. *npj Digit. Med.* 5, 3 (2022) (2022). <https://doi.org/10.1038/s41746-021-00548-8>
- [2] Hayley Denyer, J Antoni Ramos-Quiroga, Amos Folarin, Carolina Ramos, Petra Nemeth, Andrea Bilbow, Euan Woodward, Susannah Whitwell, Ulrich Müller-Sedgwick, Henrik Larsson, et al. 2022. ADHD Remote Technology study of cardiometabolic risk factors and medication adherence (ART-CARMA): a multi-centre prospective cohort study protocol. *BMC psychiatry* 22, 1 (2022), 813.
- [3] Katie L. Druce, William G. Dixon, and John McBeth. 2019. Maximizing Engagement in Mobile Health Studies: Lessons Learned and Future Directions. *Rheumatic Disease Clinics of North America* 45, 2 (2019), 159–172. doi:10.1016/j.rdc.2019.01.004 Technology and Big Data in Rheumatology.
- [4] Amelia Hemmings, Helen Sharpe, Karina Allen, Heike Bartel, Iain C Campbell, Sylvane Desrivieres, Richard JB Dobson, Amos A Folarin, Tara French, Jonathan Kelly, et al. 2023. EDIFY (eating disorders: delineating illness and recovery trajectories to inform personalised prevention and early intervention in young people): project outline. *BJPsych bulletin* 47, 6 (2023), 328–336.
- [5] Rakibul Hoque and Golam Sorwar. 2017. Understanding factors influencing the adoption of mHealth by the elderly: An extension of the UTAUT model. *International Journal of Medical Informatics* 101 (2017), 75–84. doi:10.1016/j.ijmedinf.2017.02.002
- [6] Carina Kuehne, Matthew D Phillips, Sarah Moody, Callum Bryson, Iain C Campbell, Pauline Conde, Nicholas Cummins, Sylvane Desrivieres, Judith Dineley, Richard Dobson, et al. 2024. Characterising illness stages and recovery trajectories of eating disorders in young people via remote measurement technology (STORY): a multi-centre prospective cohort study protocol. *BMC psychiatry* 24, 1 (2024), 409.
- [7] Leightley D. Siddi S. et al. Matcham, F. 2022. Remote Assessment of Disease and Relapse in Major Depressive Disorder (RADAR-MDD): recruitment, retention, and data availability in a longitudinal remote measurement study. *BMC Psychiatry* 22, 136 (2022) (2022). doi:10.1186/s12888-022-03753-1
- [8] Neto E.C. Snyder P. et al. Pratap, A. 2020. Indicators of retention in remote digital health studies: a cross-study evaluation of 100,000 participants. *npj Digit. Med.* 3, 21 (2020) 21 (2020). <https://doi.org/10.1038/s41746-020-0224-8>
- [9] Yatharth Ranjan, Zulqarnain Rashid, Callum Stewart, Pauline Conde, Mark Begale, Denny Verbeeck, Sebastian Boettcher, Richard Dobson, and Amos Folarin. 2019. RADAR-Base: Open Source Mobile Health Platform for Collecting, Monitoring, and Analyzing Data Using Sensors, Wearables, and Mobile Devices. *JMIR Mhealth Uhealth* 7, 8 (01 Aug 2019), e11734. doi:10.2196/11734
- [10] Zulqarnain Rashid, Akash Roy Choudhury, Amos A Folarin, Yatharth Ranjan, Pauline Conde, Heet Sankesara, Elisa Bruno, and Richard JB Dobson. 2024. Remote Monitoring of Epilepsy Patients with Wearable Device and Smartphone. In *Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne VIC, Australia) (UbiComp '24). Association for Computing Machinery, New York, NY, USA, 96–100. doi:10.1145/3675094.3677592
- [11] Callum L. Stewart, Zulqarnain Rashid, Yatharth Ranjan, Shaoxiong Sun, Richard J.B. Dobson, and Amos A. Folarin. 2018. RADAR-base: Major Depressive Disorder and Epilepsy Case Studies. In *Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers* (Singapore, Singapore) (UbiComp '18). Association for Computing Machinery, New York, NY, USA, 1735–1743. doi:10.1145/3267305.3267540
- [12] Callum L. Stewart, Zulqarnain Rashid, Yatharth Ranjan, Shaoxiong Sun, Richard J.B. Dobson, and Amos A. Folarin. 2018. RADAR-base: Major Depressive Disorder and Epilepsy Case Studies (UbiComp '18). Association for Computing Machinery, New York, NY, USA, 1735–1743. doi:10.1145/3267305.3267540
- [13] Rashid Z, Folarin AA, Zhang Y, Ranjan Y, Conde P, Sankesara H, Sun S, Stewart C, Laiou P, and Dobson RJ. [n. d.]. RADAR-base Platform: Digital phenotyping of mental and physical conditions through remotely collected wearables and smartphone data. *JMIR Mental Health* ([n. d.]). doi:10.2196/51259
- [14] Pratap A. Folarin A.A. et al. Zhang, Y. 2023. Long-term participant retention and engagement patterns in an app and wearable-based multinational remote digital depression study. *npj Digit. Med.* 6, 25 (2023) (2023). <https://doi.org/10.1038/s41746-023-00749-3>